

Multiple Choice (10 marks)

1. Local caching of files is common in distributed file systems, but it has the disadvantage that
 - (a) Temporary inconsistencies among views of a file by different machines can result
 - (b) The file system is likely to be corrupted when a computer crashes
 - (c) A much higher amount of network traffic results
 - (d) Caching makes file migration impossible
2. Of the following, which gives the best upper bound for the value of $f(N)$ where f is a solution to the recurrence $f(2N + 1) = f(2N) = f(N) + \log N$ for $N \geq 1$, with $f(1) = 0$?
 - (a) $O(\log N)$
 - (b) $O(N \log N)$
 - (c) $O(\log N) + O(1)$
 - (d) $O((\log N)^2)$
3. Applications developed by programming languages like _____ and _____ have this common buffer-overflow error.
 - (a) C, Ruby
 - (b) Python, Ruby
 - (c) C, C++
 - (d) Tcl, C#
4. The stack is memory for storing
 - (a) Local variables
 - (b) Program code
 - (c) Dynamically linked libraries
 - (d) Global variables
5. Of the following problems concerning a given undirected graph G , which is currently known to be solvable in polynomial time?
 - (a) Finding a longest simple cycle in G
 - (b) Finding a shortest cycle in G
 - (c) Finding ALL spanning trees of G
 - (d) Finding a largest clique in G

True / False (10 marks)

Answer True or False

6. True or False: A session symmetric key between two parties is used only once for a specific communication session.
7. True or False: Message integrity ensures that data arrives at the receiver exactly as it was sent, without alteration during transmission.
8. True or False: Windows 7 is widely regarded as having buffer-overflow vulnerabilities similar to those found in UNIX systems.
9. True or False: The AH Protocol provides source authentication and data integrity but does not ensure privacy.
10. True or False: Church's thesis was first proven by Alan Turing, establishing a definitive link between computability and Turing machines.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Write a function to remove particular data type elements from the given tuple.
12. Write a function to separate and print the numbers and their position of a given string.

End of Paper